`std::future<ResultType>::get()` or `std::shared_future<ResultType>::get()`, which retrieves the exception stored.

D.4.5. std::async function template

`std::async` is a simple way of running self-contained asynchronous tasks to make use of the available hardware concurrency. A call to `std::async` returns a `std::future` that will contain the result of the task. Depending on the launch policy, the task is either run asynchronously on its own thread or synchronously on whichever thread calls the `wait()` or `get()` member functions on that future.

Declaration

```
enum class launch
{
    async,deferred
};

template<typename Callable,typename ... Args>
future<result_of<Callable(Args...)>::type>
async(Callable&& func,Args&& ... args);

template<typename Callable,typename ... Args>
future<result_of<Callable(Args...)>::type>
async(launch policy,Callable&& func,Args&& ... args);
```

Preconditions

The expression `INVOKE(func,args)` is valid for the supplied values of `func` and `args`. `Callable` and every member of `Args` are `MoveConstructible`.

Effects

Constructs copies of `func` and `args...` in internal storage (denoted by `fff` and `xyz...` respectively).